Random Variable and Probabiilty Distribution

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Outline

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2. Probability Distribution
3. Mathematical Expectation and Variance
4. Some Common Discrete Probability Distributions
5. Some Common Continuous Probability Distributions
6. Approximating Probabilities using the Normal Distribution

Random Variable

Introduction

A variable which takes numerical values resulting from random experiments, is called a random variable (RV).

- There is a probability associated with each possible values
- Random variables are denoted by capital letters such as X, Y, Z etc.
- Possible values are denoted by small letters such as x, y, z etc.
- Example:
  - Height of students
  - Number of heads when tossing a coin three times

Types of Random Variable

- **Discrete random variable**: A random variable defined over a discrete sample space
  - Number of students in a class, sample space, $S = \{0, 1, 2, ..., \infty\}$
  - Number of correct answers in among 50 questions, $S = \{0, 1, 2, ..., 50\}$
- **Continuous random variable**: A random variable defined over a continuous sample space
  - Monthly income, $S = \{X : 0 \leq x < \infty\}$
  - Monthly profit, $S = \{X : -\infty < x < \infty\}$

Probability Distribution

Probability Distribution

- Probability distribution means the distribution of the probabilities among the different values of a random variable
- For example, when tossing a coin twice, the sample space, $S = \{HH, HT, TH, TT\}$
- Let an RV, $X =$ number of heads in two coin tosses, then X can take the values: $0, 1, 2$
- Then the probability of each value of X:

x	$P(X = x)$
0	$1/4$
1	$2/4$
2	$1/4$

Types of Probability Distributions

Depending on the type of variable, distributions are of two types:

- **Discrete probability distribution**: probability distribution of a discrete random variable
- **Continuous probability distribution**: probability distribution of a continuous random variable

Discrete probability distribution

- The probability distribution of a discrete random variable is a table, graph, formula, or other device used to specify all possible values of a discrete random variable along with their respective probabilities
- If we let the discrete probability distribution be represented by the function $p(x)$, then $p(x) = P(X = x)$ is the probability of the discrete random variable X to assume a value x
- $p(x)$ is called a probability mass function (PMF)

Probability Mass Function (PMF)

A probability mass function gives the probability of each of the value of a discrete random variable.

A function, $p(x)$, of a discrete random variable X will be called a PMF, if and only if all the following conditions are satisfied:

1. $p(x) \geq 0; \forall x$
2. $\sum_{x} p(x) = 1$
3. $P(X = a) = p(a)$

Example: PMF

In page 3, the probability mass function of the variable X representing the number of heads when a coin is tossed twice is given. It is an example of a PMF.

x	$P(X = x)$
0	$1/4$
1	$2/4$
2	$1/4$

Probabiilty Density Function (PDF)

The probability distribution of a continuous random variable is called continuous probability distribution. The probabilities are given by the probability density function (PDF) of that distribution.

A function $f(x)$ of a continuous random variable X is called the PDF of X if and only if the following conditions are satisfied:

1. $f(x) \geq 0; \forall x$
2. $\int_x f(x)dx = 1$
3. $P(a \leq X \leq b) = \int_a^b f(x)dx$

Here, $\int_x f(x)dx$ means integration of $f(x)$ over the whole sample space of X. Sometimes, it's also written as $\int_{-\infty}^{\infty} f(x)dx$.

PDF: Example

Suppose X is a continuous random variable whose density function is given by:

$$f(x) = \begin{cases} C(4x - 2x^2) & \text{if } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

1. What is the value of C?
2. Find $P(X > 1)$.

PDF: Example Solution

1.

$$\int_x f(x)dx = 1$$
$$\Rightarrow \int_0^2 C(4x - 2x^2)dx = 1$$
$$\Rightarrow C\left[4x^2/2 - 2x^3/3\right]_0^2 = 1$$
$$\Rightarrow C = 3/8$$

2.

$$P(X > 1) = \int_1^2 \frac{3}{8}(4x - 2x^2) = 1/2$$

Cumulative Distribution Function (CDF)

The cumulative distribution function (CDF) gives the probability that a random variable X is lower than some value x. Sometimes, it is also called "distribution function".

- For discrete random variable, it is given by $P(X < x) = \sum_{k=-\infty}^{x} p(k)$
- For continuous case, it is denoted by $F(x)$

$$F(x) = P(X < x) = \int_{-\infty}^{x} f(x)dx$$

- It is a non-decreasing function

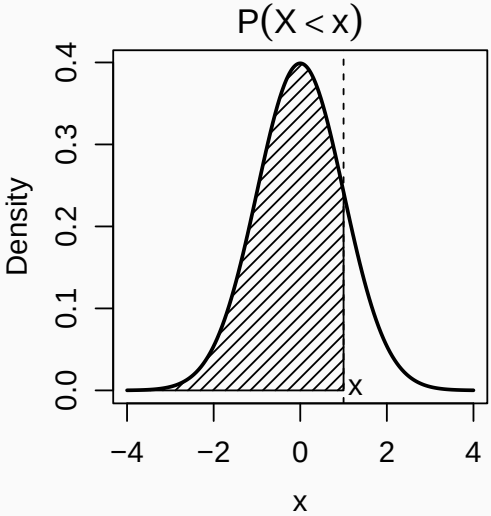
Here, summing/integrating from $-\infty$ means that the lower limit of the sum/integral is the smallest value in the domain of the random variable X.

Graphical Representaion of CDF

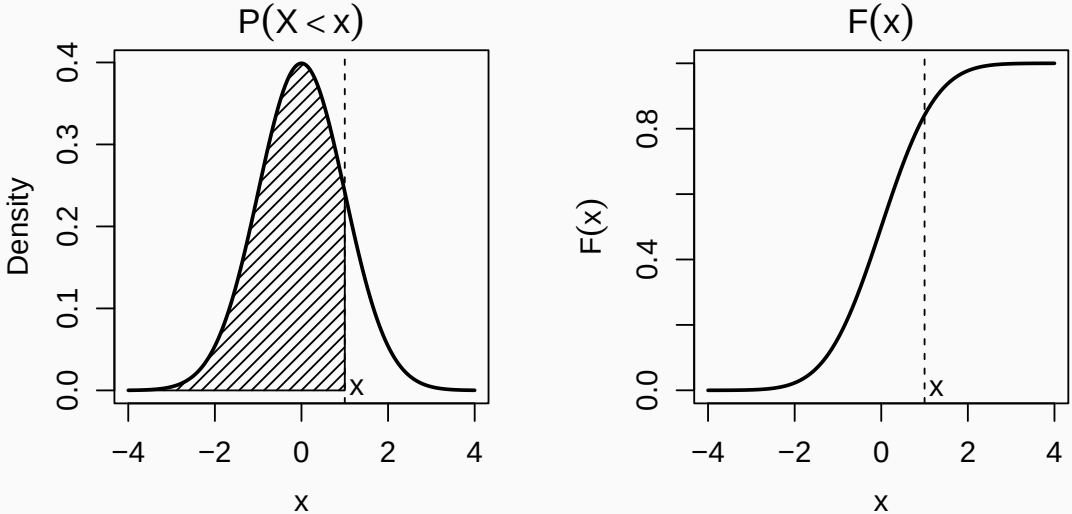
In the plot, X is a continuous random variable, it is randomly distributed (bell shaped curve).

The shaded region highlights the probability that the random variable X is less than some value x (in the plot, this value is shown to be 1 as an example).

Therefore, the highlighted area is the probability that X is less than 1.



Graphical Representaion of CDF (cont.)



Probability at a single Point for Continuous Random Variables

Let X be a continuous random variable with probability density function $f(x)$. Probability over an interval is defined as

$$P(a \leqslant X \leqslant b) = \int_a^b f(x)\,dx.$$

Probability at a single point becomes

$$P(X = a) = \int_a^a f(x)\,dx = 0.$$

- The integral over an interval of zero width (a to a) is equal to zero
- Therefore, a continuous random variable assigns no probability mass to any single value
- Probability is non-zero over intervals of width greater than zero

Why $P(X \leqslant a)$ is equal to $P(X < a)$ for Continuous Random Variables

Since $P(X = a) = 0$, adding or removing a single boundary point does not change probability.

$$\begin{aligned} P(X \leqslant a) &= P(X < a) + P(X = a) \\ &= P(X < a)\,; \quad \text{(because } P(X = a) = 0\text{)} \end{aligned}$$

Similarly,

$$P(X \geqslant a) = P(X > a)$$

For Continuous RV: $P(a \leqslant X \leqslant b) = P(a < X < b)$

Consider the probability between two values a and b:

$$P(a \leqslant X \leqslant b)$$

Expanding,

$$P(a \leqslant X \leqslant b) = P(a < X < b) + P(X = a) + P(X = b)$$

Since

$$P(X = a) = P(X = b) = 0$$

we obtain

$$P(a \leqslant X \leqslant b) = P(a < X < b)$$

CDF: Example 1

In page 3, the probability mass function of the variable X representing the number of heads when a coin is tossed twice is given.
Below, its CDF is also given:

x	$P(X = x)$	$P(X \leq x)$
0	$1/4$	$1/4$
1	$2/4$	$3/4$
2	$1/4$	1

CDF: Example 2

Suppose X is a continuous random variable whose density function is given by:

$$f(x) = \begin{cases} \frac{3}{8}(4x - 2x^2) & \text{if } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

Find the CDF of X:

$$\begin{aligned} F(X) &= \int_0^x f(x)dx \\ &= \int_0^x \frac{3}{8}(4x - 2x^2)dx \\ &= \frac{3}{8}\left[(4x^2/2 - 2x^3/3)\right]_0^x = \frac{3}{8}\left[(2x^2 - 2x^3/3) - 0\right] = \frac{3}{8}(2x^2 - 2x^3/3) \end{aligned}$$

Finding Probability from CDF

For Continuous RV:

- $P(X \leqslant a) = P(X < a) = F(a)$
- $P(X = a) = 0$
- $P(X \geqslant a) = P(X > a) = 1 - F(a)$
- $P(a < X < b) = P(a \leqslant X \leqslant b) = F(b) - F(a)$

For Discrete RV:

- $P(X \leqslant a) = F(a)$
- $P(X < a) = F(a) - P(X = a)$
- $P(X > a) = 1 - F(a)$
- $P(a < X \leqslant b) = F(b) - F(a)$
- $P(a \leqslant X \leqslant b) = F(b) - F(a) + P(X = a)$

Example: Finding Probability from CDF

Mathematical Expectation and Variance

Mathematical Expectation

Expectation of an RV, X is the weighted average of the possible values of X. The weight of each value of X is its corresponding probability.

- When X is discrete: $\mathrm{E}(X) = \sum_x x \cdot p(x)$
- When continuous: $\mathrm{E}(X) = \int_x x \cdot f(x) dx$
- The sum/integral is over all values of X
- It is often denoted by μ, that is: $\mathrm{E}(X) = \mu$
- It is also known as expected value or population mean

Variance of a Random Variable

The variance of an RV, X measures the dispersion in the values of X. It is denoted by $\mathrm{Var}(X)$ or $\mathrm{V}(X)$.

$$
\begin{aligned}
\mathrm{V}(X) &= \mathrm{E}[(X - \mathrm{E}(X))^2] \\
&= \mathrm{E}[X^2 - 2X \cdot \mathrm{E}(X) + [\mathrm{E}(X)]^2] \\
&= \mathrm{E}(X^2) - 2E(X) \cdot \mathrm{E}(X) + [\mathrm{E}(X)]^2 \\
&= \mathrm{E}(X^2) - 2[\mathrm{E}(X)]^2 + [\mathrm{E}(X)]^2 \\
&= \mathrm{E}(X^2) - [\mathrm{E}(X)]^2
\end{aligned}
$$

Here, $\mathrm{E}(X^2)$ is given by $\sum_x x^2 \cdot p(x)$ or $\int_x x^2 \cdot f(x)\, dx$ for discrete and continuous case respectively.

Expectation and Variance: Example 1

In page 3, the PMF of the random variable X representing the number of heads in two throws of a fair coin is given.
X can take the values 0, 1 and 2 with probabilities $1/4$, $1/2$ and $1/4$ respectively.
The expected value of X is:

$$\mathrm{E}(X) = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} = 0 + \frac{1}{2} + \frac{1}{2} = 1$$

The variance of X is:

$$\begin{aligned}\mathrm{V}(X) &= \mathrm{E}(X^2) - [\mathrm{E}(X)]^2 \\ &= [0^2 \cdot \frac{1}{4} + 1^2 \cdot \frac{1}{2} + 2^2 \cdot \frac{1}{4}] - 1^2 \\ &= [0 + \frac{1}{2} + 1] - 1 = \frac{1}{2}\end{aligned}$$

Expectation and Variance: Example 2

Find the expected value and the variance of the RV X whose PDF is given in page 18.

$$
\begin{aligned}
\mathrm{E}(X) &= \int_0^2 x \cdot \frac{3}{8}(4x - 2x^2)dx \\
&= \frac{3}{8}\int_0^2 (4x^2 - 2x^3)dx \\
&= \frac{3}{8}\left[4x^3/3 - 2x^4/4\right]_0^2 \\
&= \frac{3}{8}\left[4 \times 2^3/3 - 2 \times 2^4/4 - 0\right] = 1
\end{aligned}
$$

Finding the variance is left as an exercise.

Properties of Expecation and Variance

Let a, b and C be some constants, and X and Y be two independent random variables.

1. $\mathrm{E}(c) = c$
2. $\mathrm{E}(cX) = c\,\mathrm{E}(X)$
3. $\mathrm{E}(c + X) = c + \mathrm{E}(X)$
4. $\mathrm{E}(aX + bY) = a\,\mathrm{E}(X) + b\,\mathrm{E}(Y)$

1. $\mathrm{V}(c) = 0$
2. $\mathrm{V}(cX) = c^2\,\mathrm{V}(X)$
3. $\mathrm{V}(c + X) = \mathrm{V}(X)$
4. $\mathrm{V}(aX + bY) = a^2\,\mathrm{V}(X) + b^2\,\mathrm{V}(Y)$

If X and Y are dependent:
$\mathrm{V}(aX + bY) = a^2\,\mathrm{V}(X) + b^2\,\mathrm{V}(Y) + 2ab\,\mathrm{CoV}(X, Y)$.

Here, $\mathrm{CoV}(X, Y)$ is the covariance of X and Y. More on covariance in a future lecture.

Properties of Expecation and Variance: Example 1

Suppose X is a random variable with $\mathrm{E}(X) = 5$ and $\mathrm{V}(X) = 2$.

- $\mathrm{E}(2X) = 2\,\mathrm{E}(X) = 2 \times 5 = 10$
- $\mathrm{V}(2X) = 2^2\,\mathrm{V}(X) = 4 \times 2 = 8$
- $\mathrm{E}(2X - 3) = 2\,\mathrm{E}(X) - 3 = 2 \times 5 - 3 = 10 - 3 = 7$
- $\mathrm{V}(2X - 3) = 2^2\,\mathrm{V}(X) = 4 \times 2 = 8$

Some Common Discrete Probability Distributions

Bernoulli Process

When a random process or experiment, called a trial, can result in only one of two mutually exclusive outcomes, such as dead or alive, sick or well, full-term or premature, the trial is called a Bernoulli trial.

- Each trial results in one of two possible, mutually exclusive, outcomes. One of the possible outcomes is denoted (arbitrarily) as a success, and the other is denoted a failure
- The probability of a success, denoted by p, remains constant from trial to trial. The probability of a failure, $1 - p$, is denoted by q
- The trials are independent; that is, the outcome of any particular trial is not affected by the outcome of any other trial

Bernoulli Distribution

Let X be the number of successe in a single Bernoulli trial. Then X follows a Bernoulli distribution. It is written as:

$$X \sim \text{Bernoulli}(p).$$

The probability mass function (PMF) of X is given as:

$$p(x) = p^x (1-p)^{1-x}\,; \quad x = 0, 1$$

- Expected value, $\mathrm{E}(X) = 1 \times p + 0 \times (1-p) = p$
- Variance:

$$\begin{aligned}\mathrm{V}(X) = \mathrm{E}(X^2) - [\mathrm{E}(X)]^2 &= (1^2 \times p + 0^2 \times (1-p)) - p^2 \\ &= p - p^2 = p(1-p)\end{aligned}$$

Binomial Distribution

Let X be the number of successes in n Bernoulli trials, where the probability of success is p. Then the random variable X follows a binomial distribution. It is written as:

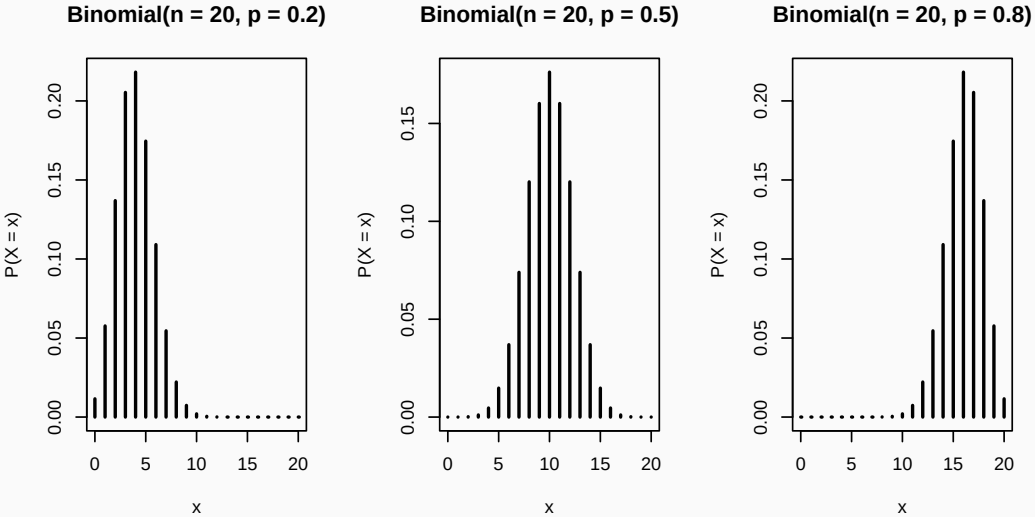
$$X \sim \mathrm{binom}(n, p).$$

The probability mass function (PMF) of X is given as:

$$p(x) =^n \mathrm{C}_x \; p^x (1-p)^{n-x} \; ; \quad x = 0, 1, 2, 3...$$

- $\mathrm{E}(X) = np$
- $\mathrm{V}(X) = np(1-p)$

Binomial Distribution: Shape



Examples of Binomial Variables

- Number of defective products among a batch of products
- Number of patients carrying a ceratain pathogen among a group of patients
- Number of heads when a coin is tossed several times

Binomial Distribution: Example

There are 3 multiple choice questions in a MCQ test. Each MCQ consists of four possible choices and only one of them is correct. If an examinee answers those MCQ randomly (without knowing the correct answers)

1. What is the probability that exactly any two of the answers will be correct?
2. What is the probability that at least two of the answers will be correct?
3. What is the probability that at most two of the answers will be correct?
4. What will be the average or expected number of correct answers?
5. Also, find the standard deviation of number of correct answers.

Binomial Distribution: Example Solution

Each MCQ has 4 options. So, when answering randomly, probability of correct answer, $p = 1/4 = 0.25$. Let the random variable X denote the number of correct answers. Then, $X \sim \text{binomial}(3, 0.25)$.

1. Probability of two correct answers, $P(X = 2) = {}^{3}\text{C}_{2}\ 0.25^{2}\,(1 - 0.25)^{3-2}$
2. Probability of *at least* two correct answers $= P(X = 2) + P(X = 3)$
3. Probability of *at most* two correct answers $= P(X = 0) + P(X = 1) + P(X = 2)$
4. Expected number of correct answers, $\text{E}(X) = 3 \times 0.25 = 0.75$
5. Variance of number of correct answers, $\text{V}(X) = 3 \times 0.25 \times (1 - 0.25)$

The calculations for question 2 and 3, are left as exercises.

Poisson Distribution

A Poisson random variable is a discrete random variable that counts the number of times a certain event occurs within fixed interval of time, or area, or volume or any other unit.

The Poisson distribution describes the probability of these counts.

Let $X \sim \text{Poisson}(\lambda)$, then the PMF of X is given by:

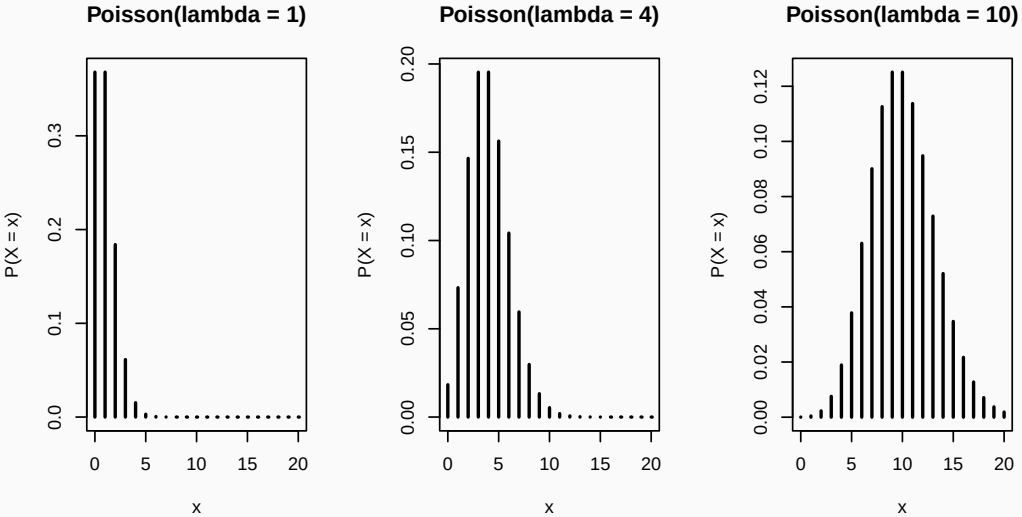
$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}; \quad \lambda > 0; \quad x = 0, 1, 2, 3, ...$$

- λ is the parameter of the distribution
- $\text{E}(X) = \lambda$
- $\text{V}(X) = \lambda$

Examples of Poisson Random Variables

- Number of phone calls received by a call center in one minute
- Number of typing errors on a page of text
- Number of accidents at an intersection in a day
- Number of emails arriving in an hour

Poisson Distribution: Shape



Assumptions of a Poisson Process

- Events occur independently of each other
- The probability of more than one event in a very small interval is negligible
- The average rate of occurrence λ is constant over the interval
- The probability of an event occurring is proportional to the length of the interval

Poisson Distribution: Example

The average number of errors on a page of a certain magazine is 0.2. What is the probability that:

1. a randomly selected page contains 0 (zero) error?
2. a randomly selected page contains 2 or more errors?
3. two consecutive pages have 3 errors?
4. What is the average error per page?
5. Also, find standard deviation of the number of errors in a page.

Poisson Distribution: Example Solution

Here, error rate, $\lambda = 0.2$.

1. Probability that a randomly selected page contains 0 errors is

$$P(X = 0) = \frac{e^{-0.2}\, 0.2^0}{0!}$$

2. Probability that a randomly selected page contains 2 or more errors is

$$\begin{aligned} P(X \geqslant 2) &= 1 - P(X < 2) \\ &= 1 - [P(X = 0) + P(X = 1)] \end{aligned}$$

The above calculations are left as an exercise.

Poisson Distribution: Example Solution (cont.)

3. Probability that two consecutive pages have 3 errors:
   The error rate for two pages is, $\lambda_{\text{new}} = 0.2 \times 2 = 0.4/$
   Therefore, the desired probability is:

$$\frac{e^{-0.4}\, 0.4^3}{3!}.$$

4. Average error per page is $\mathrm{E}(X) = \lambda = 0.2$.
5. Standard deviation of number of errors in a page, $\mathrm{SD}(X) = \sqrt{\mathrm{V}(X)} = \sqrt{0.2}$.

Some Common Continuous Probability Distributions

Uniform Distribution

A random variable with equal probability for all the possible values it can take, is called a uniform random variable.

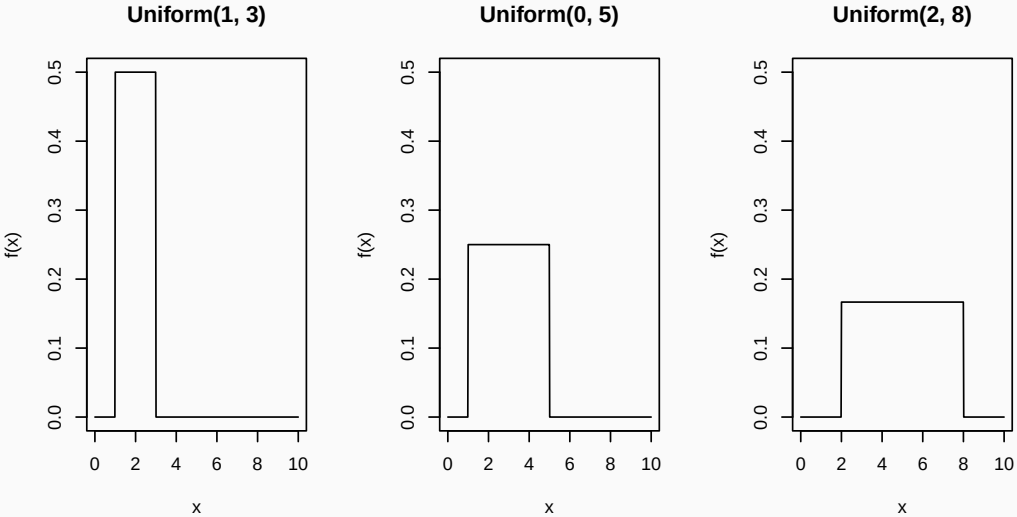
Let X be a uniformly distributed random variable in $[a, b]$, it is written mathematically as $X \sim \mathrm{uniform}(a, b)$.

The PDF of X is given by:

$$f(X) = \frac{1}{b-a}; \quad a \leqslant x \leqslant b$$

- $\mathrm{E}(X) = \frac{1}{b-a}$
- $\mathrm{V}(X) = \frac{(b-a)^2}{12}$

Uniform Distribution: Shape



Uniform Distribution: Example

The waiting time (in minutes) for train is uniform$(10, 50)$. Find the following:

1. The probability that you have to wait at least 20 minutes
2. Average waiting time
3. Standard deviation of waiting time

Uniform Distribution: Example Solution

Here, wait time, $X \sim \text{uniform}(10, 50)$

1. Probability of waiting at least 20 minutes:

$$P(X > 20) = P(20 \leqslant X \leqslant 50) = \int_{20}^{50} \frac{1}{50 - 10}$$
$$= \left[\frac{x}{40}\right]_{20}^{50} = 30/40$$

2. Average waiting time, $\mathrm{E}(X) = \frac{10+50}{2} = 30$

Question 3 is left as an exercise.

Exponential Distribution

Let X be a continuous random variable measuring the waiting time until an event occurs. The event is assumed to occur randomly but at a constant average rate. And let λ be the average number of events in one unit of time.

Then X is said to follow the exponential distribution with rate parameter λ, written as $X \sim \mathrm{exponential}(\lambda)$.

The PDF of X is given by

$$f(x) = \lambda e^{-\lambda x}\,; \quad x > 0, \quad \lambda > 0.$$

- $E(X) = 1/\lambda$, it denotes the average waiting time between events
- $\mathrm{V}(X) = 1/\lambda^2$

Examples of Exponentially Distributed Random Variables

- Waiting time until the next customer arrives at a service counter
- Time between two consecutive phone calls arriving at a call center
- Lifetime of an electronic component assuming a constant failure rate
- Time until a radioactive atom decays
- Waiting time until the next bus arrives when arrivals occur randomly
- Time between failures of a machine operating under steady conditions

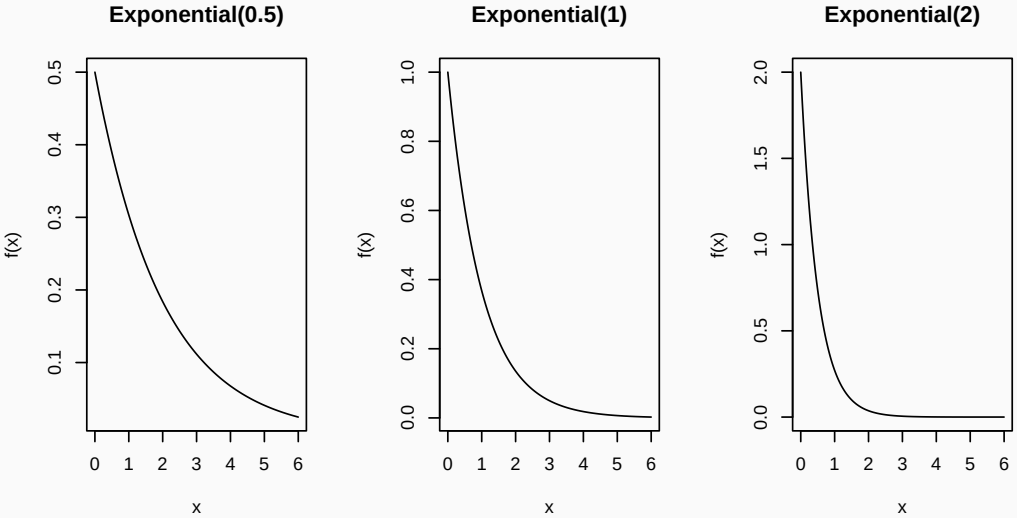
Exponential Distribution and Poisson Distribution

If events occur randomly and independently with but at a constant average rate, λ, then:

- The number of events in a fixed time interval follows a Poisson distribution
- The waiting time between those events follows an exponential distribution

Here, λ is the number of events that occur in one unit of time.

Exponential Distribution Shape



Exponential Distribution: Example 1

Average time required to repair a machine is 0.5 hours. What is the probability that the next repair will take more than 2 hours?

Let, X = time required to repair the machine
Average time required to repare a machine:

$$\mathrm{E}(X) = \frac{1}{\lambda} a = 0.5 \Rightarrow \lambda = 2$$

Therefore, $X \sim \mathrm{exponential}(2)$
Now, Probability that the next repair will take more than two hours:

$$P(X > 2) = \int_0^\infty 2e^{-2x} dx = \left[-e^{-2x} \right]_0^\infty = \left[-0 + e^{2\times 2} \right] = e^{-4} = 0.0183.$$

Normal Distribution

A continuous random variable whose distribution is symmetric and bell shaped around its mean is called a normal random variable.

Let X be a normally distributed random variable with mean μ and variance σ^2. It is written as

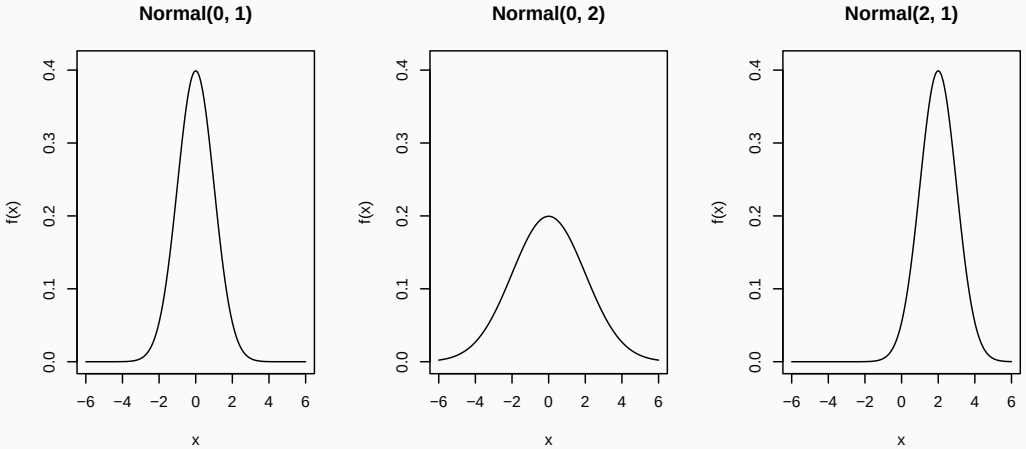
$$X \sim \mathrm{N}(\mu, \sigma^2)$$

The probability density function of X is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp(\frac{(x-\mu)^2}{2\sigma^2})\,; \quad -\infty < x < \infty$$

- $\mathrm{E}(X) = \mu$
- $\mathrm{V}(X) = \sigma^2$

Normal Distribution: Shape



Properties of Normal Distribution

- The distribution is symmetric around the mean μ
- Mean, median and mode are equal
- The spread of the distribution depends on the standard deviation σ
- Larger σ produces a wider and flatter curve
- Total area under the curve is equal to 1

Area under the Curve: Rule of Thumb

Standard Normal Distribution

A special case of the normal distribution occurs when the mean is 0 and the standard deviation is 1.

If $X \sim \mathrm{N}(\mu, \sigma^2)$, then the standardized variable

$$Z = \frac{X - \mu}{\sigma}$$

follows the standard normal distribution.

The standard normal distribution is written as

$$Z \sim \mathrm{N}(0, 1)$$

Probabilities for normal variables are usually obtained using standard normal tables.

Standard Normal Variable

For a normal random variable $X \sim \mathrm{N}(\mu, \sigma^2)$, probabilities are usually calculated using the standard normal distribution.

Any normal variable can be converted into a standard normal variable using

$$Z = \frac{X - \mu}{\sigma}$$

After converting X to Z, the probability is obtained from the standard normal table (Z-table).

The Z-table provides the cumulative probability: $P(Z \leqslant z)$, which represents the area under the standard normal curve to the **left** of z.

Finding Probabilities Using Z-table

Once the value of Z is obtained, probabilities can be computed using simple rules.

- Probability to the left of a value:

$$P(Z < z)$$

- Probability to the right of a value:

$$P(Z > z) = 1 - P(Z < z)$$

- Probability between two values:

$$P(a < Z < b) = P(Z < b) - P(Z < a)$$

These probabilities correspond to areas under the standard normal curve.

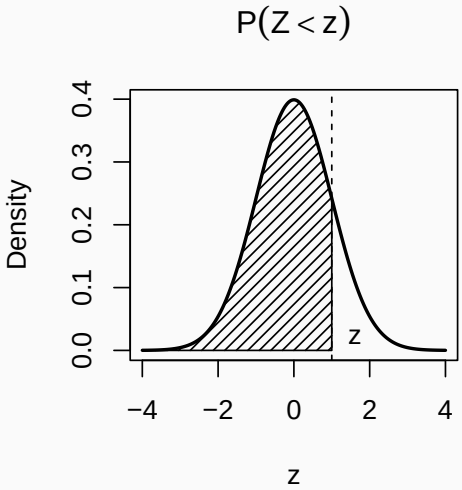
Examples of Probability Regions

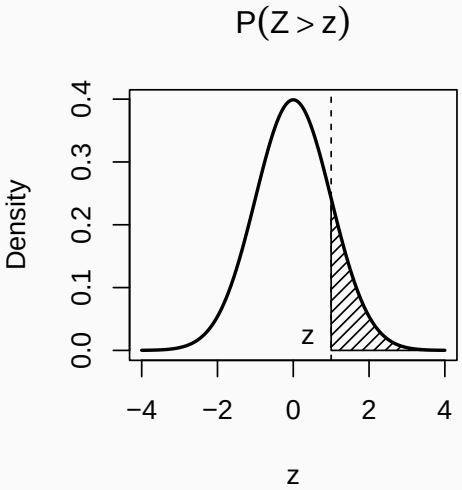
The following types of probability regions commonly appear when working with the normal distribution.

- Probability that a variable is less than a value: $P(Z < z)$
- Probability that a variable is greater than a value: $P(Z > z)$
- Probability that a variable lies between two values: $P(a < Z < b)$
- Probability that a variable lies outside an interval: $P(Z < a \cup Z > b)$

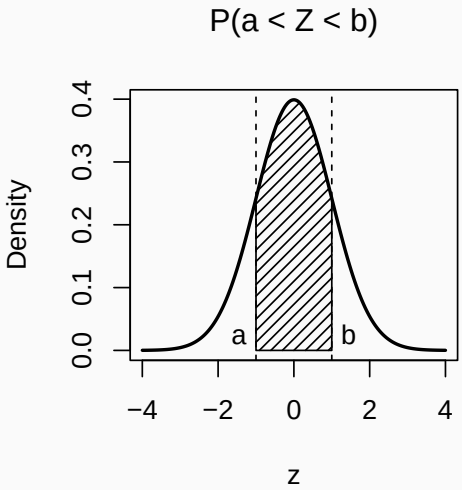
In the next slides we will visualize these probabilities using shaded regions on the normal curve.

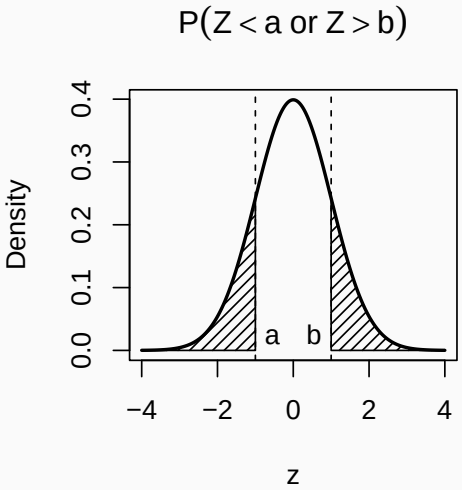
Normal Distribution: Probability Regions





Normal Distribution: Probability Regions (cont.)





Normal Distribution: Example

The heights of students in a university are normally distributed with mean 170 cm and standard deviation 10 cm.

Find the following:

1. The probability that a randomly selected student is taller than 180 cm
2. The probability that a student's height lies between 160 cm and 180 cm
3. The average height of students

Normal Distribution: Example Solution

Let height $X \sim \mathrm{N}(170, 10^2)$

1. Probability that a student is taller than 180 cm:

$$Z = \frac{180 - 170}{10} = 1$$

$$P(X > 180) = P(Z > 1) = 0.1587$$

2. Probability that height lies between 160 and 180:

$$Z_1 = \frac{160 - 170}{10} = -1$$

$$Z_2 = \frac{180 - 170}{10} = 1$$

$$P(160 < X < 180) = P(-1 < Z < 1) = 0.6826$$

Approximating Probabilities using the Normal Distribution

Thank you.